UNIVERSITY EXAMINATION

Web Services, Principles, Patterns and Constraints
Back-End Web Development

Time Allowed:	2 Hours 30 Minutes	**Total Marks:**	100
Instructions:	Answer ALL questions in Section A and ANY THREE in Section B		

SECTION A — Short Answer Questions (40 Marks)

Answer ALL questions in this section. Each question carries the marks indicated.

1. Define a **Web Service** and explain how it differs from a standard web application. **[3 marks]**
2. State THREE motivations for using web services in distributed systems, drawing on the concept of interoperability. **[3 marks]**
3. Distinguish between a **URI** (Universal Resource Identifier) and a **URL** (Uniform Resource Locator), giving an example of each. **[4 marks]**
4. List and briefly explain the FIVE design constraints of REST. **[5 marks]**
5. What does **HATEOAS** stand for, and what role does it play in a RESTful web service? **[3 marks]**
6. Identify the FOUR Richardson Maturity Model levels (Level 0 – Level 3) and state the key characteristic of each level. **[4 marks]**
7. Explain what **over-fetching** and **under-fetching** are in the context of REST, and name the alternative architectural style that addresses these issues. **[4 marks]**
8. State THREE properties that every service in a Service-Oriented Architecture (SOA) must have, giving a one-sentence explanation of each. **[3 marks]**
9. List FOUR SOA principles and explain how they promote loose coupling. **[4 marks]**
10. Name the THREE architecture styles covered in the notes and write one distinguishing feature of each. **[3 marks]**
11. Outline the parts of a URL using the following template: **scheme://host:[port]/path[?query-string][#anchor]** **[4 marks]**
12. Name TWO tools and TWO technologies used when working with Web APIs. **[2 marks]**

SECTION B — Structured Questions (60 Marks)

Answer ANY THREE questions. Each question carries 20 marks.

Question 13 — Web APIs and Service-Oriented Architecture [20 marks]

(a) Explain the concept of a Web API. In your answer, describe the role of the HTTP protocol and how it facilitates stateless client-server communication. **[5 marks]**

(b)	Service-Oriented Architecture (SOA) encourages breaking an application down into smaller, independent software components. With the aid of an example, describe how this philosophy is applied in a real-world system such as an e-commerce platform.	**[6 marks]**
(c)	Each service in SOA must be: (i) Well-defined, (ii) Self-contained, and (iii) A blackbox. Discuss what each property means for a service that handles user authentication.	**[5 marks]**
(d)	List and explain FOUR principles that govern web service design and development.	**[4 marks]**

Question 14 — REST and Architectural Styles [20 marks]

(a)	REST stands for Representational State Transfer. Describe REST as an architectural style and explain the claim made by RESTful web services regarding scalability and generality of component interfaces.	**[5 marks]**
(b)	Explain the following REST design principles with examples: (i) Simplicity (ii) Extensibility (iii) Scalability (iv) Distributed hypermedia	**[8 marks]**
(c)	In REST design patterns, it is important to properly track long-running operations. Describe how this is achieved, including how identifiers are used and how clients track progress using GET requests.	**[4 marks]**
(d)	Explain **optimistic locking** in the context of concurrent updates in RESTful services and why it is necessary given the statelessness constraint.	**[3 marks]**

Question 15 — SOAP, gRPC and Comparing Architectural Styles [20 marks]

(a)	SOAP (Simple Object Access Protocol) is used in distributed environments. Describe the three components of a SOAP-based system (service requester, service provider, service registry) and the role each plays.	**[6 marks]**
(b)	gRPC is described as a high-performance, open-source universal RPC framework. Explain how gRPC allows clients to invoke remote services as if they were local objects, and describe the role of Protocol Buffers (ProtoBuf) in this process.	**[6 marks]**
(c)	Compare REST and gRPC across the following dimensions, providing a brief explanation for each: (i) Protocol used (ii) Message format (iii) Supported HTTP verbs (iv) Learning curve	**[6 marks]**
(d)	State ONE advantage and ONE disadvantage of SOAP over REST.	**[2 marks]**

Question 16 — Front-End Design and Web Standards [20 marks]

(a)	Explain what the Document Object Model (DOM) is. In your answer, describe its tree-like structure and explain why it is language-independent and abstract.	**[5 marks]**
(b)	The three layers of a web front-end are: Structure, Presentation, and Behaviour. For each layer, name the technologies used and describe the role the layer plays in a web application.	**[6 marks]**

(c)	Web standards exist to help developers build accessible and compatible applications. Describe the GOALS of web standards as per the W3C, and explain why compatibility across devices is important.	**[5 marks]**
(d)	A front-end developer must understand how to design front-end apps by analysing project structure. Explain how the DOM serves as the core of a front-end project and how JavaScript (JS/TS) is used to extend DOM functionality.	**[4 marks]**

Question 17 — Application and Design Scenario [20 marks]

A software company is building a hospital management system. The system needs to integrate a patient records service, an appointment scheduling service, and a pharmacy service, all of which will be consumed by mobile, web, and smart device clients.

(a)	Recommend an architectural style (REST, SOAP, or gRPC) for this system. Justify your choice by referring to at least THREE criteria relevant to this scenario (e.g. interoperability, performance, device support).	**[6 marks]**
(b)	Using SOA principles, describe how the system should be decomposed into services. Identify each service, explain its discrete task, and describe how the services should communicate with each other.	**[6 marks]**
(c)	Design a set of RESTful endpoints for the **Appointment Scheduling** service. Your answer should include appropriate HTTP verbs, resource URIs, and a brief description of what each endpoint does. Provide at least FOUR endpoints.	**[5 marks]**
(d)	Explain how the system should handle the problem of **service interoperability** when the mobile client (using JSON) needs to communicate with a legacy desktop system that uses XML.	**[3 marks]**

— END OF EXAMINATION —